Do Rules of Thumb Move with Prices?

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9th November 2026

Abstract

Many drivers buy fuel in round amounts such as £20. Using credit card fuel purchases in Great Britain, 2016–2025, I study how this rule of thumb responds to prices. Rounding changes little with prices, but drivers quickly revise the round amount: its within-card price elasticity is 0.62. Rounding therefore changes liters per fill far less than a fixed nominal budget would. At a fixed amount, the next fill comes sooner by 55 percent of the liters lost, relative to other fills. Models that treat simple rules as fixed may overstate how much they distort purchases.

*Rice University, Jones Graduate School of Business. benedictgk@rice.edu. I am grateful for feedback from Devin Pope, Emir Kamenica, Karthik Srinivasan, Matt Notowidigdo, Richard Thaler, Thomas Covert, Walter Zhang, faculty and students at Chicago Booth, participants at the Russell Sage Foundation Behavioral Economics Summer Institute and the Behavioral Science & Policy Association Conference, and Ivo Wengraf and Steve Gooding from the RAC Foundation. I thank Fable Data for sharing data for this research, and their staff for helpful discussions. The views expressed are the author's and do not necessarily reflect the views of Fable Data. Fable Data had the right to review this paper before its dissemination to ensure that it did not contain confidential information.

1 Introduction

Few prices are as visible to households as the price of fuel, which moves inflation expectations, sentiment, news coverage, and votes (Coibion and Gorodnichenko, 2015; Binder, 2018; Arezki et al., 2022; Binder and Makridis, 2022; Kim and Yang, 2022; Cummings et al., 2024). Drivers see that price every time they fill up, yet many purchase by a rule of thumb, or decision heuristic, stopping the pump at a round amount such as £20. Unlike most goods, which are sold in discrete units, fuel is divisible to the penny. A purchase of a round amount therefore reflects the driver's own rule rather than how fuel is sold. How that rule behaves when prices change is not known.

The rule can be read in two ways. First, a round amount can be a fixed nominal budget for each fill, a form of mental budgeting (Thaler, 1985, 1999; Heath and Soll, 1996). The driver spends £20 whatever the price, so a higher price buys fewer liters (the budget constraint alone makes quantity fall, as in Becker (1962)) and brings the next fill forward, as press reports suggested when prices rose in 2022 (Bloomberg News, 2022). Drivers often reason in amounts per fill, as in Shaquille O'Neal's advice that an empty tank costs $80 to fill but a half tank only $20 (Greene, 2018; Last Week Tonight with John Oliver, 2022). Second, the driver can keep stopping at a round number but pick a larger one when fuel costs more, so a price change shows up in the amount paid per fill rather than in liters or timing. In the terms of Kőszegi and Matějka (2020), the first reading is a nominal budget, under which the elasticity of the amount paid with respect to the price is zero. The second implies a positive elasticity, which reaches one under a quantity budget, where the driver fixes the liters.

I test whether drivers use the rule of thumb more or less often as prices change, whether and how quickly they change the round amount they stop at, and how rounding changes the response of liters per fill. I use credit card transaction data covering 4.7 million fuel purchases paid at the pump with over half a million credit cards in Great Britain from 2016 to 2025. A pay-at-the-pump payment buys only fuel, which makes its amount the purchase. Following the same cards over time separates changes in behavior from changes in who is in the data. Of these purchases, 37 percent end at, or a penny or two above, a multiple of £5. The period spans a large, reversing price change: the monthly average petrol (gasoline) price rose from 102p a liter (about $5.00 a US gallon at a fixed $1.30 per £1) in early 2016 to 188.5p (about $9.30) in July 2022 and then fell by 30 percent by June 2025.

Drivers keep rounding but revise the round amount. Whether they round at all moves little with prices. The share of round fills drifted down and did not recover when prices fell. Within card, a 10 percent higher price raises the round amount by about 6 percent, and the within-card average tracks the price month by month through the 2022 spike and its

reversal. Most of the response arrives in the month of the price change, as cards step up the £5 grid after rises and down after falls. The elasticity lies between the two benchmarks, at about 0.62, 0.65 when consumer prices replace the time trend, and 0.51 without any trend on the 2022–25 fall alone, when the petrol price fell while other consumer prices rose.

Because the round amount is revised, rounding changes the liters response per fill much less than a fixed amount would. Liters per round fill fall with the price by 0.11 log points more per log point than liters per other fill, against 0.74 if the amount were never revised.

When drivers hold the amount fixed, a higher price buys fewer liters, and the next fill comes forward by less than the liters lost. To measure this, I compare a card's round fills with its other fills, using the time to the card's next fill at any station, paid at the pump or in the store, so that any response of driving to the price common to both kinds of fill cancels. Of the liters lost, 55 percent shows up as an earlier fill. For round fills that repeat the previous amount, the share is 85 percent, and full pass-through cannot be rejected. A fixed amount thus works at most partly as a budget for the time between fills, since the shortfall can also reflect how full the tank is when drivers fill and refill, the driving ahead, and fills the card does not see. The amount elasticity is similar for round fills paid in the store and across income proxies, car dependence, fill frequency, age, and region. The timing difference is negative in every group, and reweighting cards to the adult population's age and region shares leaves both nearly unchanged.

These results contribute to three literatures. The first documents round numbers and other heuristics in household decisions. People set goals at round numbers (Pope and Simonsohn, 2011; Allen et al., 2017), employers pay wages at them (Dube et al., 2025), home buyers' offers cluster at them (Pope et al., 2015; Mateen et al., 2026), and borrowers request round loan amounts (Hu and Kraft, 2026). Credit cardholders repay round amounts (Sakaguchi et al., 2026) and match repayments to balances (Gathergood et al., 2019), and car buyers target monthly loan payments (Argyle et al., 2020). This evidence largely treats these rules as static. Savers choose retirement contribution rates at multiples of 5 percent, and Benartzi and Thaler (2007) suggest that when the maximum allowed rate rose, some switched from one heuristic to another. NFT traders who list at round prices revise them less often but in larger steps (Rosen and Spaenjers, 2026). For fuel, Lynn et al. (2013) document round amounts in 1,301 purchases at one convenience store over a few days in 2005. Following more than a million purchases on the same cards through a large price cycle, I show that the rule stays in use while the round amount it produces is revised quickly and in both directions.

The second studies mental accounting. Hastings and Shapiro (2013) show that households' choice of gasoline grade responds far more to gasoline prices than to equivalent changes

in income, consistent with a separate mental account for gasoline. With almost all petrol in Great Britain of one grade, a driver who holds the amount fixed cannot downgrade and must buy fewer liters or refill sooner. Kőszegi and Matějka (2020) model mental budgets, set in nominal spending or in quantities, as a way to simplify choice when shocks are costly to attend to, and many people report keeping budgets (Zhang et al., 2022). A round amount may also be an adaptive reference point (Kőszegi and Rabin, 2006; Thakral and Tô, 2021); I study it as a rule for choosing an action, the margin the card data observe. Drivers use a nominal amount per fill, revise it quickly when prices move, and when they hold it fixed it works at most partly like a budget for the timing of the next fill. The fuel price's visibility should make updating cheap. Quick updating is also a natural response if people expect fuel prices to persist, as survey evidence suggests they do on average (Anderson et al., 2013).

The third studies how consumer demand responds to prices, in fuel with high-frequency or purchase-level data (Levin et al., 2017, 2022; Gelman et al., 2023) and, more generally, how behavioral rules shape the demand curve (Becker, 1962). Approximately half of the downward slope of Lyft riders' demand occurs as the price drops below a dollar value (List et al., 2023), and used-car prices fall discontinuously as the odometer crosses a round number of miles (Lacetera et al., 2012; Kraft and Rao, 2026). Purchases need not track use: buyers of storable goods time purchases around sales (Hendel and Nevo, 2006), and drivers buy ahead of announced tax rises (Coglianese et al., 2017). Fuel demand also splits into distance driven and on-road fuel economy (Knittel and Tanaka, 2021). I separate the margin at which the response happens, the liters bought at each fill or the time to the next, which demand estimates based on spending or volumes alone cannot distinguish. Because drivers revise the round amount, liters per fill fall far less than under a fixed nominal budget.

2 Data, Predictions, and Empirical Design

2.1 Institutional Details and Data

Setting. In Great Britain, pump prices are posted per liter on signs visible from the road, include all taxes, and typically end in .9 pence. Prices move mostly with the price of crude oil, set on world markets, together with the exchange rate and refining margins (Competition and Markets Authority, 2023), which makes the price variation I use national and set largely outside the country. Drivers fill their own tanks, and with no clip to hold the nozzle open, they hold the trigger throughout. To stop at a round amount, they must watch the display and release the trigger at the right moment. The display shows both the amount spent and the liters bought, so a driver who stops at a round number of pounds chooses the nominal

unit.

Fuel types. Unlike in the United States (Hastings and Shapiro, 2013), almost all drivers in Great Britain buy one grade of petrol. By a 2021 government estimate, the premium grade was about 5 percent of sales (Department for Transport, 2021), so switching grade is a minor margin and a minor source of error in liters. I do not observe whether a car runs on petrol or diesel. About a third of cars in my setting run on diesel (Society of Motor Manufacturers and Traders, 2024), whose price generally moves with petrol, and cars cannot switch between the two. Measuring the price as two-thirds petrol and one-third diesel leaves both main results nearly unchanged (Appendix Table A1).

Card transactions data. Paying by card removes one ordinary reason to buy a round amount, avoiding coins in change. I use anonymized transactions on credit cards in Great Britain from 2016 to June 2025 from Fable Data (2026), which collects card data from several issuers. Fable credit card spending closely tracks official series (Gathergood and Guttman-Kenney, 2021; Grigoli and Pugacheva, 2024). Household financial transaction data track official gasoline sales closely in the United States as well (Baker, 2018). I do not include data for Northern Ireland because people there may cross the border to pay in euros. A card is not a person: a household may use several cards, and fills paid with cash or another card are not seen. To limit unseen fills, I keep regular-use cards, those with an average of at least five non-fuel card payments a month over their time in the sample. Months without payments count as zero, and because the criterion is set once per card, changes in card use over time do not move cards in or out of the sample. The criterion keeps 44 percent of credit cards with a fuel payment, which account for 63 percent of fuel payments. Relative to the adult population, cardholders in my sample are concentrated at ages 30–49, and their regional distribution matches the population apart from lower coverage in London (Appendix Figure A1).

Fills. A pay-at-the-pump payment (merchant category code 5542, "automated fuel dispensers", or payment processor "pay at pump") buys only fuel. A *fill* is a pay-at-the-pump payment of at least £1 or another petrol-station payment of at least £10 (merchant category codes 5541, "service stations (with or without ancillary services)", and 9752, "UK petrol stations, electronic hot file"). The £10 minimum avoids counting small store payments as fills, as in Gelman et al. (2023) and Hofmann and Reinmuth (2026). Payments are dated by transaction date. Days to the next fill are measured to the card's next day with any fill, paid at the pump or in the store, and gaps of more than 60 days are dropped as likely missed fills.

Round and other fills. A fill is *round* if it is a multiple of £5 or one or two pence above one, and its round amount is that multiple. Pumps stopped on a target often overshoot by a penny or two, more so at higher prices. Between 2019 and 2022, exact multiples of £5 fell from 28 to 18 percent of fills while amounts one or two pence over rose from 14 to 16 percent (Appendix Figure A2). A fill is *other* if it is more than 10 pence from a multiple of £5 and is not a whole number of pounds (or a penny or two over one), so it is unlikely to be aimed at a round amount. Remaining fills (near misses within 10p of a multiple and whole-pound amounts off the £5 grid) may be aimed at a target and are left out of the comparisons.[1]

Prices and quantity. I use public data on the daily average national petrol pump price from the RAC Foundation (2026), which tracks the government's weekly series (Department for Energy Security and Net Zero, 2025, correlation 0.999). I estimate the liters bought as the amount paid divided by this price.

2.2 Predictions and Empirical Design

Theoretical predictions. A driver buys Q liters of fuel at price P and pays $A = PQ$. Under a fixed nominal budget per fill, A does not move with P, so the elasticity $d\log A/d\log P$ is 0. Under a fixed quantity per fill, a quantity budget in the terms of Kőszegi and Matějka (2020), A moves one-for-one with P, so the elasticity is 1.

Now hold the amount fixed, so that a round fill's liters fall one-for-one with the price. Suppose the driver refills when the tank is back at the level it was at this fill and uses $c(P)$ liters a day. Days to the next fill are then $G = Q/c(P)$, which gives $d\log G/d\log P = -1 - d\log c/d\log P$ for a round fill. When fuel prices are higher, a fixed amount buys fewer liters, and the next fill comes sooner unless the driver drives less. If the driver instead refills at a lower tank level than at this fill, the gap also draws on fuel already in the tank, so days respond less than one-for-one to the liters bought even with no change in driving.

Let δ^G and δ^Q be the round–other differences in the price elasticities of days to the next fill and of liters per fill, respectively. If, after both kinds of fill, the driver next refills at the tank level just before this fill and driving responds to the price in the same way, then $\delta^G = \delta^Q$, and the liters lost show up in full as an earlier next fill. At the other extreme, $\delta^G = 0$. Relative to other fills, a round fill then lasts the same number of days whatever it buys. The amount works as a budget for time, and the driver drives less or refills at a lower tank level. The test requires that need at the fill, meaning the tank level and the

[1] Of single fills followed by another fill within 60 days, 10 percent are near misses and 8 percent other whole-pound amounts. Counting every fill within 10p of a multiple as round, at that multiple, leaves both main results nearly unchanged (Appendix Table A1).

driving ahead, does not move with the price within a card's fills of one amount. Otherwise a more expensive fill could last longer because the tank was fuller or less driving lay ahead. Section 3.3 examines this.

Estimation. Let t index the day of card i's fill. For the amount, I regress the log round amount (never zero; Chen and Roth, 2024) on the log price that day:

$$\log A_{it} = \alpha_i + \beta \log P_t + \gamma \tau_t + \phi_{\ell(t)} + \mu_{m(t)} + \varepsilon_{it}, \tag{1}$$

with card fixed effects α_i, a linear time trend τ_t, an effect $\phi_{\ell(t)}$ for the months of England's national COVID-19 lockdowns (March–June 2020, November 2020, and January–March 2021), and month-of-year effects $\mu_{m(t)}$. Under a fixed nominal budget $\beta = 0$, and under a fixed quantity $\beta = 1$. The trend absorbs general growth in nominal amounts but also part of the long-run movement in the price, so I report β with and without it. For timing,

$$\log G_{it} = \alpha_{i,k(it)} + \beta^o \log P_t + \delta^G R_{it} \log P_t + \gamma_{R_{it}} \tau_t + \phi_{R_{it},\ell(t)} + \lambda_{R_{it},w(t)} + \mu_{R_{it},m(t)} + e_{it}, \tag{2}$$

where R_{it} indicates a round fill, β^o is the price elasticity of days to the next fill after other fills, $\lambda_{R_{it},w(t)}$ are day-of-week effects, and $\alpha_{i,k(it)}$ are card × amount fixed effects, with one cell for each round amount the card buys and one for its other fills. The trend coefficient γ, the lockdown effect ϕ, and the day-of-week and month-of-year effects all carry the subscript R_{it}. Each therefore differs between round and other fills, and the point estimates equal those from separate regressions for each kind of fill. Within a cell, a round fill's liters fall one-for-one with the price. Round fills' liters elasticity is thus exactly -1, and δ^Q is -1 minus other fills' liters elasticity.

The same regression with log liters per fill as the outcome gives δ^Q, and with $\log(Q/G)$ it gives $\delta^Q - \delta^G$. A version with calendar-month fixed effects in place of the trend, lockdown indicator, and month-of-year effects absorbs anything that affects both kinds of fill in a given month, such as driving, lockdowns, or seasons. In this version β^o is identified only from within-month price changes, and I report δ^G from it.

Standard errors in these regressions are clustered by card and by month. Because the price varies only over time, I also estimate monthly versions that treat the months as the observations: within-card month effects (for timing, the month-specific round–other difference) regressed on the monthly log price, with heteroskedasticity- and autocorrelation-robust standard errors (Lazarus et al., 2018).

Estimation samples. Most fuel bought on credit cards is paid in the store. Of card-days with a fill, 74 percent have a single fill paid in the store, and a few have more than one. A store payment can include other goods, and two fills on one day cannot be told apart. The main observations are therefore card-days with exactly one fill, paid at the pump. The round-share analysis (Section 3.1) uses all 4,656,318 of these card-days, on 552,766 cards. Each month's share of round fills is estimated from all these fills with card fixed effects, so each purchase counts once, and the monthly series is regressed on the monthly price. Because the price is national and the share drifts slowly, inference treats months, not fills, as the observations. The amount analysis uses the 1,583,567 round fills of the 218,422 cards with at least two. The timing analysis compares round and other fills on the same card. It uses cards with at least three of each that are followed by another fill within 60 days: 1,425,150 fills on 52,620 cards, with a median of 8 days between fills. Fills alone in their card $\times$ amount cell drop out of the main timing estimates. On all cards the timing difference is similar (Appendix A).

A broader sample adds round fills paid in the store. Fuel bought with other goods lands on a round amount only by chance, since a driver would have to pick the goods to round the total. A round store payment is thus very likely fuel alone, while a store payment at any other amount may include goods and its liters are unknown. On card-days with a single store fill and no pump payment there are 3,339,691 round fills. In the timing analysis, 414,980 of them join the round fills, with their own card $\times$ amount cells, and the other fills remain those paid at the pump. The pump sample stays the main one because it is fuel alone by construction and supplies the timing test's comparison group. Section 3 reports both results on this broader sample and on cards that pay at the pump for at least 90 percent of their fills.

3 Results

3.1 Whether to Round Moves Little with the Price

Pay-at-the-pump amounts bunch at multiples of £5 (Figure 1, Panel A). Within card, the share of round fills fell steadily, from 44 percent in 2016 to 31 percent in 2025 (Appendix Figure A3). Because the price also rose over much of this period, the coefficient on the log price is negative without a trend (-0.29, Table 1, Panel A, column 2), but it disappears once a linear trend is allowed (-0.03, column 1). Over the 2016–22 price rise, that estimate's 95 percent confidence interval runs from a fall of 6.8 percentage points in the share to a rise of 3.6. After 2022 the share did not recover as prices fell (Appendix Table A2), so whether

drivers round moves little with the price.

Precision does move with the price. From 2021 to 2022 the share of fills within 10p of a £5 multiple barely moved while the share of those that hit it exactly fell from 49.0 to 38.9 percent (Appendix Figure A3, Panel C). At higher prices each second of pumping costs more, so the same delay in releasing the trigger likely overshoots by more.

3.2 The Round Amount Moves with the Price

The average within-card log round amount tracks the log price month by month, through the rise of 2021–22, the peak of July 2022, and the fall that followed (Figure 1, Panel B). In equation (1) with a linear trend, a 10 percent higher price raises the round amount by about 6 percent, or £1.84 at the mean round amount of £30.15 ($\beta = 0.62$; Table 1, Panel B, column 1). The estimate does not reflect a change in which cards round: on cards with round fills both before and from 2022 it is 0.63 (column 3), and the monthly series give similar estimates (columns 4–5).

Attributing the rise in amounts to the price is harder, since nominal amounts would drift up with general prices anyway. Over 2016–22 both rose, and without a trend the elasticity is 0.81 (column 2). The fall of 2022–25 separates the two. The petrol price fell by up to 30 percent while core consumer prices rose by 14 percent, so general drift pushed amounts up while petrol pushed them down. Amounts still fell with the price, with an elasticity of 0.51 and no trend (Appendix Table A1, Panel B). On the 2021–22 rise, where the drift works in the price's favor, it is 0.74. The two windows bracket the full-sample estimate, and replacing the trend with the consumer price index gives 0.65 (Panel A). I take 0.62 as the preferred estimate (Appendix A reports further checks).

A fixed nominal budget per fill ($\beta = 0$) is rejected in every specification, and a fixed quantity ($\beta = 1$) in every one except the monthly series without a trend (1.03, SE 0.09; Appendix Table A1, Panel B). Drivers keep a nominal round amount but revise it with the price, so most of a price change shows up in what they spend at each fill and the rest in the liters they buy.

One concern is that households might shift larger fills onto the credit card when fuel prices are higher, raising the within-card round amount with no change in the rule. The estimate changes little on cards that pay at the pump for at least 90 percent of their fills or with a control for the card's non-fuel payments that month. A second concern is that the price could change which fills a driver rounds, along with how much fuel is wanted. On cards that round at least 90 percent of their fills, which leaves little room for that, the estimate also changes little. Round fills paid in the store give a similar estimate to pump round fills

of £10 or more, the range store fills cover. These results, in Appendix Table A1, do not rule out shifts to cash or to other cards, which are unobserved.

The round amount responds to the price quickly, though not entirely within the month. In a distributed-lag model with 12 monthly lags, the current month's price accounts for 0.50 of a 12-month total elasticity of 0.68 (Appendix Table A1, Panel B). Day-to-day changes within a month move the amount by about half as much (0.33 with calendar-month fixed effects, Panel A), still between the two benchmarks, and holding today's price fixed, a rise over the previous eight weeks lowers the amount (Appendix Table A3). Both fit an amount that follows an average of recent prices, or station prices that lag the national average.

Drivers adjust by switching between round amounts. Between a card's consecutive round fills, a 10 percent price rise raises the probability of moving up at least one £5 step by 9 percentage points, and a 10 percent fall raises the probability of moving down by 8 points, against base rates of about a third each (Appendix Table A1, Panel C). Amounts also change when the price does not: 63 percent of consecutive round fills differ when the price has moved by less than 0.5 percent in between. These changes barely track how empty the tank is (Section 3.3), so few round amounts come from rounding the liters needed to refill (Appendix C).

Because the amount is revised, rounding changes the liters response per fill much less than a fixed amount would. With card × fill-type fixed effects, so that the amount may change (Table 1, Panel C, column 3), liters per round fill fall by only 0.36 log points per log point of price, close to $1 - \beta$, and liters per other fill by 0.26. Liters per round fill therefore fall by 0.11 (SE 0.02) more than liters per other fill, against 0.74 if the amount were never revised. The contrast compares purchases drivers chose to round with their other purchases. It describes purchasing per fill, and any price-related difference in need between the two enters it.

3.3 The Timing of the Next Fill

Holding the round amount fixed, a higher price buys fewer liters, and Figure 2 shows that the next fill then comes sooner. In Panel A, days to the next fill fall more steeply with the price after round fills than after the same cards' other fills. Panel B shows the month-by-month round–other difference, which falls with the price across the whole period, including after the 2022 peak, but less steeply than the liters a round fill buys (yellow line).

In equation (2), relative to the same card's other fills, a 1 percent higher price lowers liters per round fill by 0.74 percent but brings the next fill forward by 0.41 percent (Table 1, Panel C, column 1), so 55 percent of the liters lost shows up as an earlier fill. With the

next fill 12.3 days after a round fill on average, a 10 percent higher price brings it forward by about 0.5 days more than after other fills. For other fills, days to the next fill and liters fall with the price by about as much. Net of the trend, liters per day after other fills barely respond, in line with small short-run elasticities of fuel consumption (Hughes et al., 2008).

The estimates reject both benchmarks of Section 2.2. The next fill comes sooner than if a round fill lasted the same time whatever it buys ($\delta^G = 0$), but not as much sooner as full pass-through ($\delta^G = \delta^Q$) would imply, with $\delta^Q - \delta^G = -0.33$ (SE 0.03). If need at the fill does not move with the price within a card's fills of one amount, the shortfall reflects a budget for time, with drivers driving less or letting the tank run lower when a fill buys fewer liters, or simply a lower refill level after round fills than after other fills at any price.

The timing difference holds across specifications and samples, and so does the shortfall where Appendix Table A1 reports liters per day. Month fixed effects, which absorb shocks common to all fills such as the panic buying of September 2021 (Hofmann and Reinmuth, 2026), leave −0.42 (Table 1, Panel C, column 2). The difference is similar in the monthly series, on cards that pay at the pump for at least 90 percent of their fills (where almost no fills are dropped), when round fills paid in the store join the round group, and for whole-pound amounts off the £5 grid. Across 11 variants of the fill and gap definitions, including one that counts only pump payments as the next fill, it lies between −0.39 and −0.49 (Appendix A).

The main threat to reading the shortfall as a budget for time is selection on need. Drivers choose the amount with need in mind. Across a card's round amounts, at a given price, the elasticity of days to the next fill with respect to the amount is 0.30. With amounts unrelated to need it would be one if drivers next refilled at the tank level just before each fill, so a larger amount likely goes with an emptier tank or more driving ahead. If the same holds within one amount, a card's £20 fills bought at high prices, which buy fewer liters, would be those with less need, and the next fill would come later than the liters imply. Selection on the tank level appears small. If drivers rounded the liters they need, the amount would rise with the time since the previous fill, a proxy for how empty the tank is, but it barely does (elasticity 0.023 holding the price fixed; Appendix C gives the specification). In a simulation calibrated to that elasticity, such selection lowers the share of the liters lost that shows up as an earlier fill from 84 to about 78 percent, against 55 percent in the data (Appendix C). Controlling for the previous gap and the amount spent at the previous fill leaves −0.37 (Appendix Table A1). Neither the proxy nor the simulation captures driving planned after the fill. The repeat fills below come closest to a test of it.

Fills paid with cash or another card, which I do not see, merge two gaps. If the unseen fill's liters respond to the price like those of other fills, the measured response after a round fill is pulled toward that after other fills, which biases the estimate toward zero. I cannot

measure their share, so they may account for part of the shortfall. Two other explanations find little support. Routine weekly fills would put extra mass at 7, 14, 21, and 28 days after round fills, but the excess over neighboring days is the same after both kinds of fill. Expectations that a recent rise will reverse would bring the next fill forward, but a one-standard-deviation rise does so by less than half a percent, leaving δ^G almost unchanged (Appendix Table A3).

For the 13 percent of round fills that repeat the card's previous amount, the next fill does track the liters bought: 85 percent of the liters lost shows up as an earlier fill, and full pass-through cannot be rejected ($\delta^Q - \delta^G = -0.10$, SE 0.09; Table 1, Panel C, column 4). Repeating an amount suggests a rule rather than the liters needed, so these fills are likely less exposed to selection on need. Their shortfall is about a third of the average, though imprecisely estimated. All round fills on the same cards give 63 percent, so the higher share is not mainly about which cards repeat amounts. Drivers who keep buying the same amount may cycle the tank between similar levels, as the benchmark assumes, selection on need may explain more of the average shortfall than the tank-level simulation allows, since it omits planned driving, or fewer unseen fills may follow repeat fills. The data do not separate these. Either way, a fixed amount works at most partly as a budget for time.

3.4 Heterogeneity

Who rounds? Among 60,468 credit cards with at least six pay-at-the-pump fills in 2017–2019, the average card in the bottom decile of card spending made 53 percent of these fills at round amounts and the average card in the top decile 28 percent. Conditioning on age, region, and the card's other characteristics, a standard deviation more log card spending goes with a round share 10.5 percentage points lower (58,721 cards) and a standard deviation less area deprivation with one 2.2 points lower. Rounding is thus more common on cards with lower spending and in poorer areas, consistent with the prediction of Kőszegi and Matějka (2020) that lower-income households more often budget in nominal terms.

Who updates? Figure 3 shows little heterogeneity in the amount elasticity. The blue lines reweight cards to the adult population's age and region shares, following Baker (2018), and the headline estimates barely change (Appendix Table A1). The amount elasticity falls slightly with age, from 0.69 at 18–29 to 0.59 at 60 and over, and varies little across the regions of Great Britain. The round–other timing difference varies more, between −0.65 and −0.18 across regions, but is negative in every group.

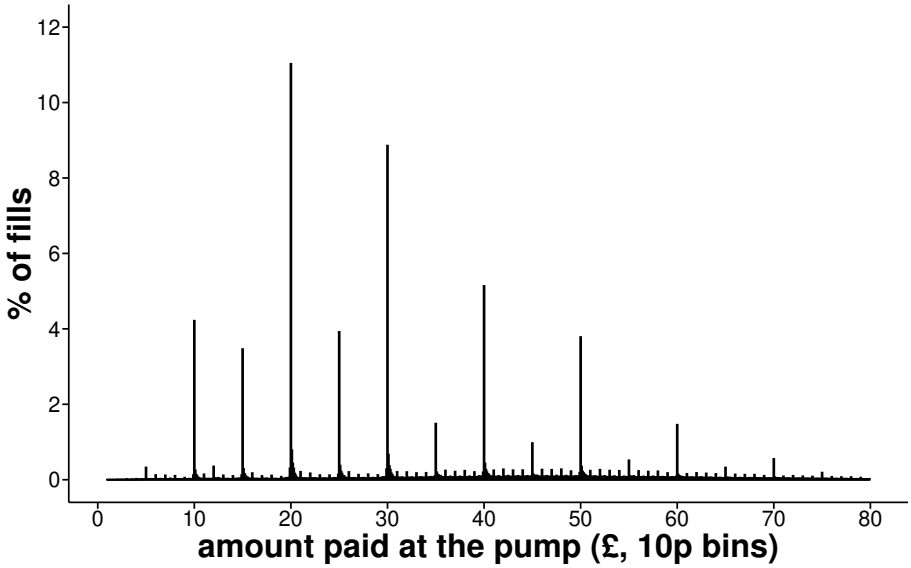
The amount elasticity also differs little with income proxies, car dependence, and how often the card buys fuel (Appendix Figure A4). Of the 34 tests of differences across these

characteristics, none survive a Bonferroni correction. Cards that rounded most of the time in their first year update less (0.58) than cards that rarely did (0.85), though the pooled test of this difference does not survive the correction either. The 2.5 percent of cards in the amount sample that keep one round amount (at least five round fills, 80 percent of them at one amount) have an elasticity of 0.24 (SE 0.02), low by construction, since they are selected on keeping their amount.

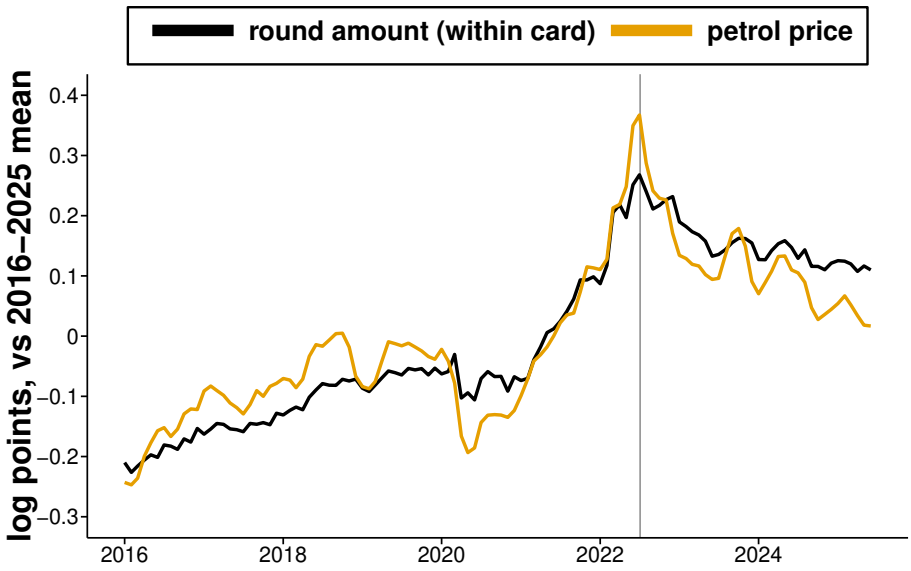
4 Conclusion

Do rules of thumb move with prices? For round-number fuel purchases, the rule barely does: drivers stop at a multiple of £5 about as often when fuel prices are high or low. The round amount they stop at does. Drivers move to a larger round amount soon after a price rise and back down after a fall, so most of a price change shows up in what they spend at each fill. Because the round amount is revised, rounding changes the liters response per fill far less than a fixed nominal budget would. When drivers keep the amount, the next fill comes sooner, relative to their other fills, by 55 percent of the liters lost in the main sample and by 85 percent for fills that repeat the previous amount. A fixed amount therefore works at most partly as a budget for time. Because the rule stays in use while the round amount it produces moves, models that treat round numbers or other simple rules as fixed may overstate how much they distort purchases and understate how quickly behavior adapts to a new price level.

Figure 1: Round Amounts Move with the Price



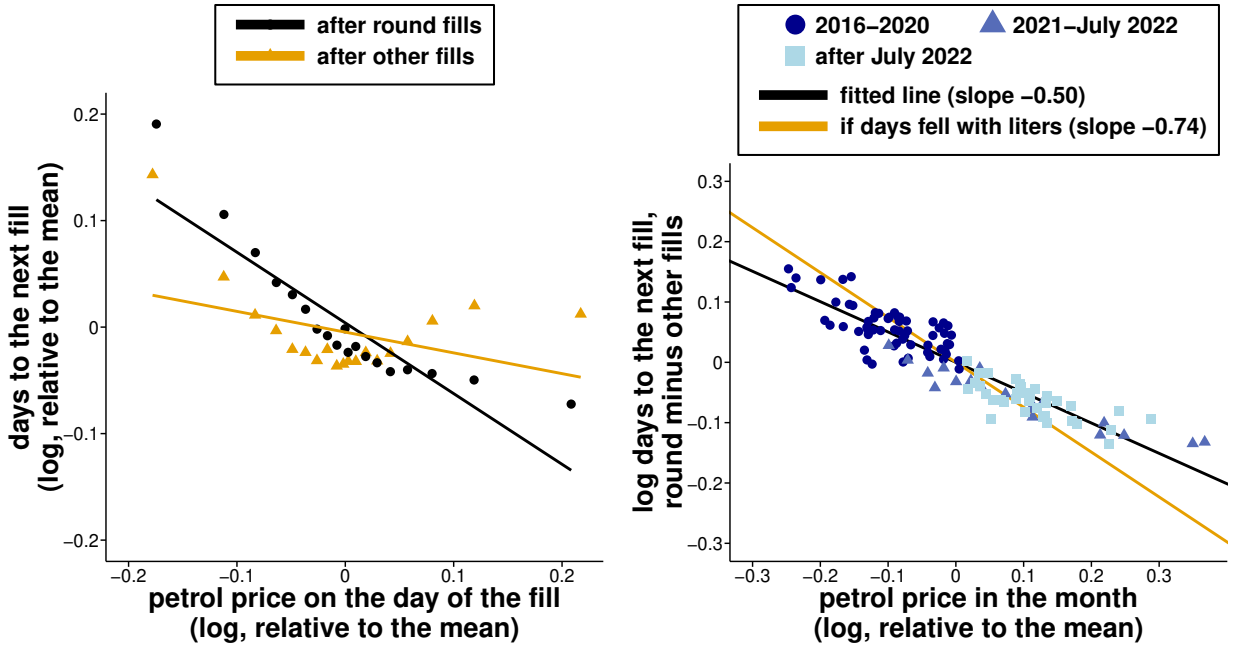
(a) Pay-at-the-pump amounts



(b) Within-card round amount and the petrol price, by month

Notes: Panel A: distribution of amounts on card-days with a single pay-at-the-pump fill. Panel B: month effects from a regression of the log round amount on card and month fixed effects, and the log monthly average petrol price, each measured from its 2016–2025 mean, on the round fills of the amount sample (Section 2.2); gray line: July 4, 2022, the record price. Regular-use credit cards.

Figure 2: Days to the Next Fill and the Price



(a) Fills, within card and amount

(b) Round − other difference, by month

Notes: Panel A: points are mean log days to the next fill in 20 quantiles of the log national petrol price on the day of the fill, both relative to the card × amount mean (round fills, black) or the card mean (other fills, yellow); lines are least-squares fits through the points. The steeper line after round fills means that the next fill comes sooner after a round fill bought at a higher price than after the same card's other fills. Panel B: points are, for each month, the difference between round and other fills in log days to the next fill (month effects from a regression with card × amount and month fixed effects), against the log monthly price, both relative to their means. Black line: least-squares fit through the months, without a trend (Appendix Table A1 reports −0.53 with month-of-year effects and −0.40 with a trend). Yellow line: slope −0.74, the round − other difference in log liters per fill (Table 1, Panel C, column 1), which days to the next fill would follow if the next fill came when the liters bought ran out. Timing sample (Section 2.2).

Table 1: Round Amounts, Liters, and the Timing of the Next Fill

Panel A. Share of round fills: coefficient on the log petrol price

	Trend	No trend	Steps
	(1)	(2)	(3)
Log price	−0.03	−0.29	−0.02
	(0.04)	(0.06)	(0.04)
Months	114	114	114

Panel B. Log round amount: elasticity with respect to the petrol price

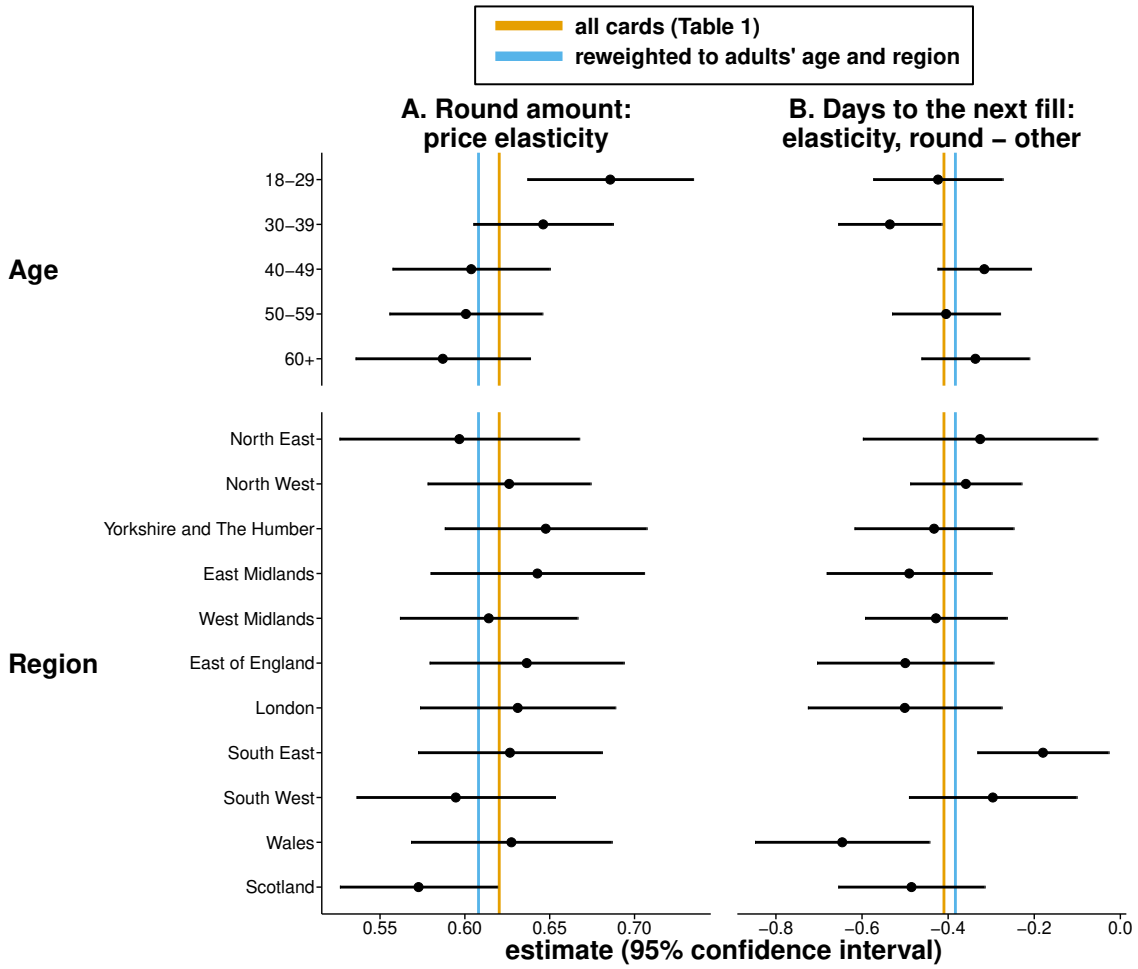
	Fills	No trend	Same cards pre and post	Monthly HAR	Monthly, 12-month diff.
	(1)	(2)	(3)	(4)	(5)
Log price	0.62	0.81	0.63	0.64	0.59
	(0.02)	(0.03)	(0.02)	(0.06)	(0.06)
Linear trend	Yes	No	Yes	Yes	–
Observations	1,583,567	1,583,567	473,888	114 months	102 months

Panel C. Round − other difference in the elasticity with respect to the petrol price

	Same amount	Same amount, month FE	Amount free	Repeat fills
	(1)	(2)	(3)	(4)
Log days to next fill	−0.41	−0.42	−0.17	−0.56
	(0.03)	(0.03)	(0.02)	(0.09)
Log liters per day	−0.33	–	0.06	−0.10
	(0.03)		(0.03)	(0.09)
Log liters per fill	−0.74	–	−0.11	−0.66
	(0.02)		(0.02)	(0.03)
Observations (fills)	1,303,925	1,303,925	1,425,150	176,475

Notes: Panel A: all card-days with a single pay-at-the-pump fill (Section 2.2); month effects from a fill-level regression of a round-fill indicator on card and month fixed effects, regressed on the log monthly price with month-of-year effects and a COVID-lockdown indicator, with a linear trend (column 1), without (column 2), or with level shifts in March 2020 and March 2022 (column 3; Appendix B); a coefficient of −0.10 means a 10 percent higher price goes with a share 1 percentage point lower. Panel B: equation (1); dependent variable the log round amount (columns 1–3: round fills; card fixed effects, COVID-lockdown indicator, month-of-year effects; column 3: cards with at least three round fills before and from 2022). Columns 4–5: within-card month effects of the log round amount regressed on the log monthly price with month-of-year effects (column 4) or in 12-month differences (column 5). Panel C: equation (2); round − other difference in the elasticity of log days to the next fill, log liters per day (liters bought ÷ days to the next fill) and log liters per fill, with a trend, COVID indicator, day-of-week, and month-of-year effects, each specific to round and other fills, or month fixed effects (column 2). Columns 1, 2, and 4: card × amount fixed effects, so each round fill is compared at a fixed amount; liters per fill is then the sum of the two estimates above it. Column 3: card × fill-type fixed effects, so the amount may change; all three rows estimated; its timing rows include moves to larger amounts, which last less than proportionally longer (Section 3.3). Column 4 uses round fills whose previous fill was the same round amount, and the other fills of cards with at least three of each. Means in levels: Panel B sample, round amount £30.15 (22.4 liters); Panel C sample, days to the next fill 12.3 after round fills and 12.5 after other fills, liters per fill 23.8 and 32.9. Standard errors in parentheses: clustered by card and month (fills); equal-weighted cosine HAR, Lazarus et al. (2018) (months).

Figure 3: Estimates by Cardholder Age and Region



Notes: Separate regressions for each group. Panel A: elasticity of the round amount with respect to the fill-day price (equation 1, with trend), amount sample (Section 2.2). Panel B: the price elasticity of days to the next fill after round fills minus that after the same cards' other fills (equation 2), timing sample; more negative means the next fill comes sooner after a round fill bought at a higher price. Bars: 95 percent confidence intervals, clustered by card and month. Yellow lines: pooled estimates (Table 1); blue lines: pooled estimates with cards reweighted to the adult population's age and region shares in 2021. Age: the cardholder's age band. Region: Great Britain, by the cardholder's postcode. Areas are matched to postcode sectors with the Office for National Statistics Postcode Directory.

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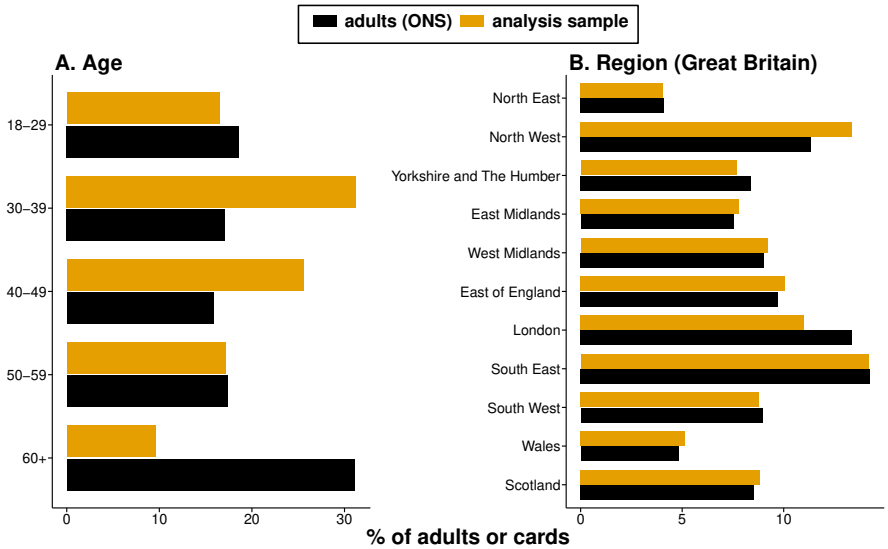
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Supplemental Appendix Accompanying *"Do Rules of Thumb Move with Prices?"*

A Robustness

Appendix Table A1 reports both main results under alternative specifications and samples, including cards reweighted to the adult population's age and region shares, which Appendix Figure A1 compares with the analysis sample. The timing difference lies between −0.39 and −0.49 across 11 variants of the fill and gap definitions: exact multiples only; fills of at least £5 or £20 at the station; the next fill defined by any station payment or by the pump only; maximum gaps of 30, 45, and 90 days; all cards; cards in use throughout the gap; and the half of cards with the steadiest gaps between fills relative to their average, among cards with at least five gaps. With a trend, which is nearly collinear with the price within a window, the window estimates are 0.41 for the fall and 0.45 for the rise. The amount responds more to rises than to falls (Appendix Table A1, Panel C), but core consumer prices rose through the fall and card fixed effects remove only a constant drift, so the difference may reflect drift rather than asymmetric updating. A full tank that stops large round targets short at low prices would raise the amount with the price, most for small tanks, yet the elasticity is 0.72 for the third of cards with the largest tanks, against 0.67 for all cards with a tank estimate (Appendix Table A1).

Figure A1: Distribution of Age and Region for Cards in Data Compared to Distribution of Adults



Notes: Shares of regular-use credit cards with a pay-at-the-pump fill, active in 2021, by cardholder age band and region, against the adult population from the Office for National Statistics (ONS) mid-2021 population estimates.

Table A1: Robustness of the Two Main Results

	Round	Round − other difference	
	amount elasticity (1)	Log days to next fill (2)	Log liters per day (3)
Panel A. Specification and sample			
Main specification (Table 1)	0.62 (0.02)	−0.41 (0.03)	−0.33 (0.03)
No trend	0.81 (0.03)	−0.46 (0.03)	
Calendar-month fixed effects	0.33 (0.10)	−0.42 (0.03)	
Round = within 10p of a £5 multiple	0.63 (0.02)	−0.39 (0.03)	−0.36 (0.03)
Reweighted to adults' age × region	0.61 (0.02)	−0.38 (0.04)	−0.34 (0.04)
Excluding 2020–2021	0.66 (0.02)	−0.43 (0.04)	
Cards paying at the pump for ≥90% of fills	0.60 (0.03)	−0.39 (0.11)	
Cards rounding ≥90% of fills	0.54 (0.03)		
Cards in the top third of tank size	0.72 (0.04)		
Cards with ≥80% of round fills at one amount		−0.40 (0.20)	−0.24 (0.20)
Whole-pound amounts off the £5 grid		−0.41 (0.07)	−0.38 (0.07)
Round fills at the pump or in the store	0.57 (0.02)	−0.37 (0.03)	−0.38 (0.04)
Round fills paid in the store only	0.54 (0.02)	−0.32 (0.04)	−0.43 (0.05)
Control: log non-fuel card payments	0.61 (0.02)		
Core CPI instead of a trend	0.65 (0.02)		
Amount and price deflated by core CPI	0.64 (0.02)		
Price: two-thirds petrol, one-third diesel	0.61 (0.02)	−0.39 (0.03)	
Controls shared by the two fill types		−0.43 (0.03)	−0.42 (0.03)
Controls: previous fill's gap and amount		−0.37 (0.03)	
Panel B. Speed of the amount response			
Distributed lag (0–12 months): month 0	0.50 (0.08)		
Distributed lag: sum of months 0–12	0.68 (0.02)		
Price rise, Jan 2021–Jul 2022 (no trend)	0.74 (0.04)		
Price fall, Jul 2022–Jun 2025 (no trend)	0.51 (0.02)		
Monthly series, HAR: no trend	1.03 (0.09)	−0.53 (0.04)	
Monthly series, HAR: with trend	0.64 (0.06)	−0.40 (0.03)	
Panel C. Moving a £5 step between consecutive round fills			
Step up, on the price rise	1.00 (0.07)		
Step down, on the price fall	−0.74 (0.04)		
Rises − falls: net step (up − down)	0.49 (0.16)		
Rises − falls: log round amount	0.21 (0.09)		

Notes: Column 1: equation (1); columns 2–3: round − other difference, equation (2). Column 3, log of liters bought ÷ days to the next fill, gives $\delta^Q - \delta^G$. Standard errors clustered by card and month; monthly series (Section 2.2): HAR. Calendar-month fixed effects: the amount elasticity comes only from price changes within a month. Within 10p: near misses count as round, at the nearest multiple. Tank: the 95th percentile of liters on the card's other fills (cards with at least 10). One main amount: at least five round fills, 80 percent of them at one amount; column 1 is blank because selection on keeping the amount biases it toward zero (Section 3.4). Whole-pound amounts off the grid replace round fills. Shared controls: one trend, lockdown indicator, and set of day-of-week and month-of-year effects for both fill types. Price windows: fill-day price, card fixed effects, month-of-year effects for the fall; starting the rise in April 2021 gives 0.70 (SE 0.04). Panel C: consecutive round fills of a card at most 120 days apart; linear probability of a higher (lower) multiple on the rise (fall) in the log price since the previous round fill; the last two rows are rise − fall differences in the net step (up − down) and in the change in the log round amount; card and month-of-year fixed effects and a COVID indicator.

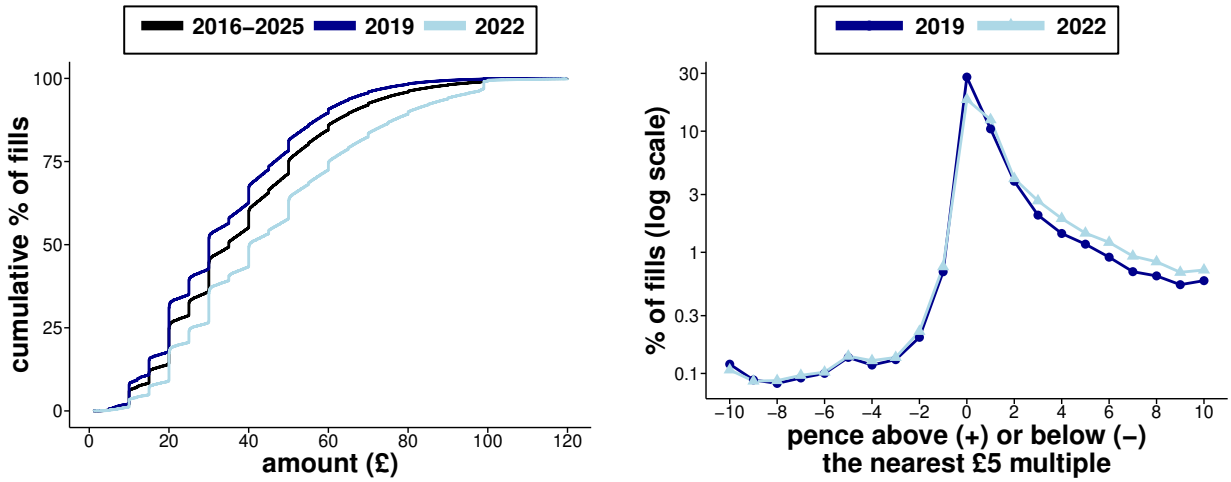
B Whether to Round

The cumulative distribution of single pay-at-the-pump amounts jumps at multiples of £5 (Appendix Figure A2, Panel A), and overshoots of a penny or two are far more common than undershoots (Panel B).

Appendix Figure A3 plots, by month, the within-card round and exact shares, attempts (amounts within 10p of a multiple), the hit rate (exact multiples as a share of attempts), and the price. In regressions of the monthly shares on the log price fitted to 2016–2021 (Appendix Table A2), the round share and attempts show no negative relation with the price once a trend is allowed, while exact multiples and the hit rate do. Adding the price to the trend improves the 2022–2025 forecasts of exact multiples and the hit rate and worsens those of attempts and the round share, whose root mean squared error rises from 1.4 to 2.4 percentage points. These forecasts omit the March 2022 step discussed next.

Within cards, the round share falls by 2.7 percentage points from the 12 months before March 2022 to the 12 months after, 1.5 more than the average change across other years' Marches ($z = -4.0$). For fuel paid in the store, the excess fall is 0.3 points ($z = -0.6$). The step coincides with a 3.0-point fall in the share of fuel paid at the pump and with pay-at-the-pump pre-authorization holds of about £100 replacing £1 card checks. Because the step is specific to paying at the pump, I do not attribute it to the price. With level shifts in March 2020 and March 2022 in place of a trend, the coefficient on the log price is −0.02 (Table 1, Panel A, column 3). If the holds also capped fills at £100, capped fills would count as round and could become more common at higher prices. In no year were more than 0.3 percent of pay-at-the-pump fills between £100.00 and £100.09. The round-amount elasticity is similar for fills paid in the store (0.54), which holds do not affect, and the rise in amounts begins in 2021 (Figure 1), before the holds.

Figure A2: Pay-at-the-Pump Amounts

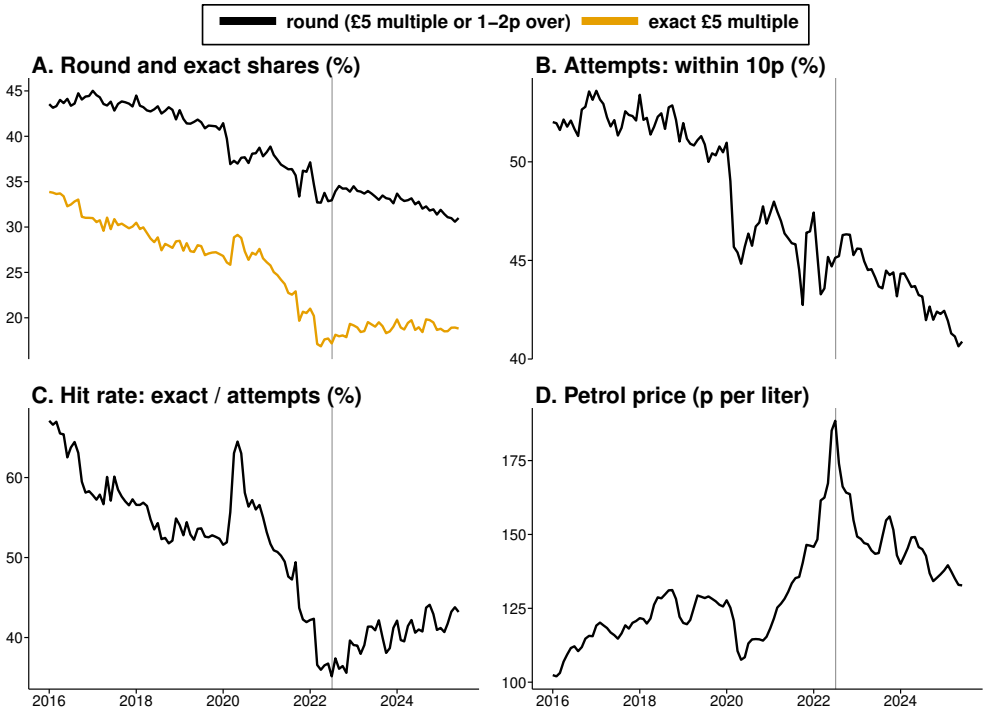


(a) Cumulative distribution of amounts

(b) Distance from the nearest £5 multiple

Notes: Card-days with a single pay-at-the-pump fill, regular-use credit cards. Panel A: cumulative share of fills by amount in 10p bins, for 2016–2025 (black), 2019 (dark blue), and 2022 (light blue). Panel B: fills a given number of pence above or below the nearest multiple of £5, as a percentage of all fills that year, 2019 and 2022, on a log scale. Round fills are those at 0, +1, and +2 pence.

Figure A3: Round Shares, Precision, and the Petrol Price



Notes: Month effects from regressions of indicators for a round fill (a multiple of £5 or 1–2p over), an exact multiple, and an attempt (within 10p of a multiple) on card and month fixed effects, pay-at-the-pump fills of regular-use credit cards; hit rate: exact multiples as a share of attempts. Gray line: July 4, 2022, the record price.

Table A2: The Share of Round Fills, Precision, and the Price

Share of fills	Log price, 2016–2021		RMSE 2022–2025 (pp)	
	No trend	Trend	Trend only	Price + trend
	(1)	(2)	(3)	(4)
Round share	−0.18	0.10	1.4	2.4
	(0.12)	(0.05)		
Exact multiples	−0.35	−0.15	2.3	0.9
	(0.07)	(0.03)		
Attempts (within 10p)	−0.12	0.15	1.0	2.1
	(0.12)	(0.05)		
Hit rate (exact ÷ attempts)	−0.69	−0.52	5.2	1.9
	(0.04)	(0.02)		

Notes: Each row's monthly within-card share (a fraction) regressed on the log monthly petrol price with month-of-year effects and a COVID-lockdown indicator; HAR standard errors. A coefficient of −0.10 means a 10 percent higher price goes with a share 1 percentage point lower. RMSE: root mean squared error of predictions for 2022–2025 from models fitted to 2016–2021.

C Selection on Need and Price Expectations

Selection on need. The tank-state elasticity in Section 3.3 (0.023) comes from regressing the log ratio of the round amount to the card's usual amount (the median of its previous five round amounts) on log days since the previous fill, with the log price, the trend, the lockdown indicator, and card, day-of-week, and month-of-year fixed effects.

To calibrate how much bias this elasticity allows, I simulate 10,000 cards from 2016 to June 2025 that burn fuel daily, refill at a level redrawn at each fill, and buy a full or partial tank or a round amount.[2] A round amount follows one of two rules: a nominal amount set from a remembered price, whatever the tank level, or the liters needed (a random share of the empty tank) times the price, both rounded to £5. With only the first rule, the estimator recovers −0.84 when the truth is −1, an attenuation from the varying refill level and missed fills. The simulated liters difference is about −1, so each estimate is also the share of the liters lost that shows up as an earlier fill. With only the second, it recovers −0.29, and the tank-state elasticity is 0.22. Shares of 10 to 15 percent of fills under the second rule bracket the data's tank-state elasticity (0.017 to 0.025, against 0.023), and the estimator then recovers

[2]Tank of 40 to 70 liters; daily use of 2 to 6 liters with price elasticity −0.05; refill level between 10 and 60 percent of capacity; a card-specific 20 to 70 percent of fills round, the rest full tanks (70 percent) or partial fills; the nominal amount 10 to 40 liters valued at an exponential average of past prices with a 15-day half-life (180 days in a check); the liters needed 50 to 100 percent of the empty space at the pump price; station prices within 4 percent of a national price that tracks the petrol price from 2010 to 2025; 3 percent of fills unobserved; the data's gap cap and sample rules.

−0.80 to −0.77. Across 16 simulated samples, the Monte Carlo standard deviation of each estimate is below 0.01. A 180-day half-life, which lowers the simulated amount elasticity from about 1 to 0.56 to 0.60 with a trend (0.62 in the data), makes every estimate about 0.01 more negative. In the simulation, this selection and the other frictions lower the share from 100 to about 78 percent, about half of the observed shortfall to 55 percent, unless days since the previous fill is a much noisier proxy for the tank level in the data than in the simulation. The simulation models selection on the tank level only, not on driving planned after the fill, which also enters the choice of amount (Section 3.3).

Price expectations. A driver who expects a recent price rise to reverse should buy fewer liters now and come back sooner, and one who expects it to continue should buy more now, as drivers do ahead of announced tax rises (Coglianese et al., 2017). Appendix Table A3 adds the change in the log national price over the previous eight weeks to the fill-level models, holding the price level fixed. The fill-size coefficients have the sign mean reversion predicts but are small: a one-standard-deviation eight-week rise (4.7 percent) lowers other fills' liters by about 0.4 percent. After round fills, relative to other fills, the next fill also comes slightly sooner (−0.09), but a one-standard-deviation rise moves it by less than half a percent. After other fills, timing does not respond (0.04). The round-amount coefficient is not a test, because the amount follows a short average of past prices. With today's price held fixed, a recent rise means a lower remembered price and a smaller amount, without any expectations.

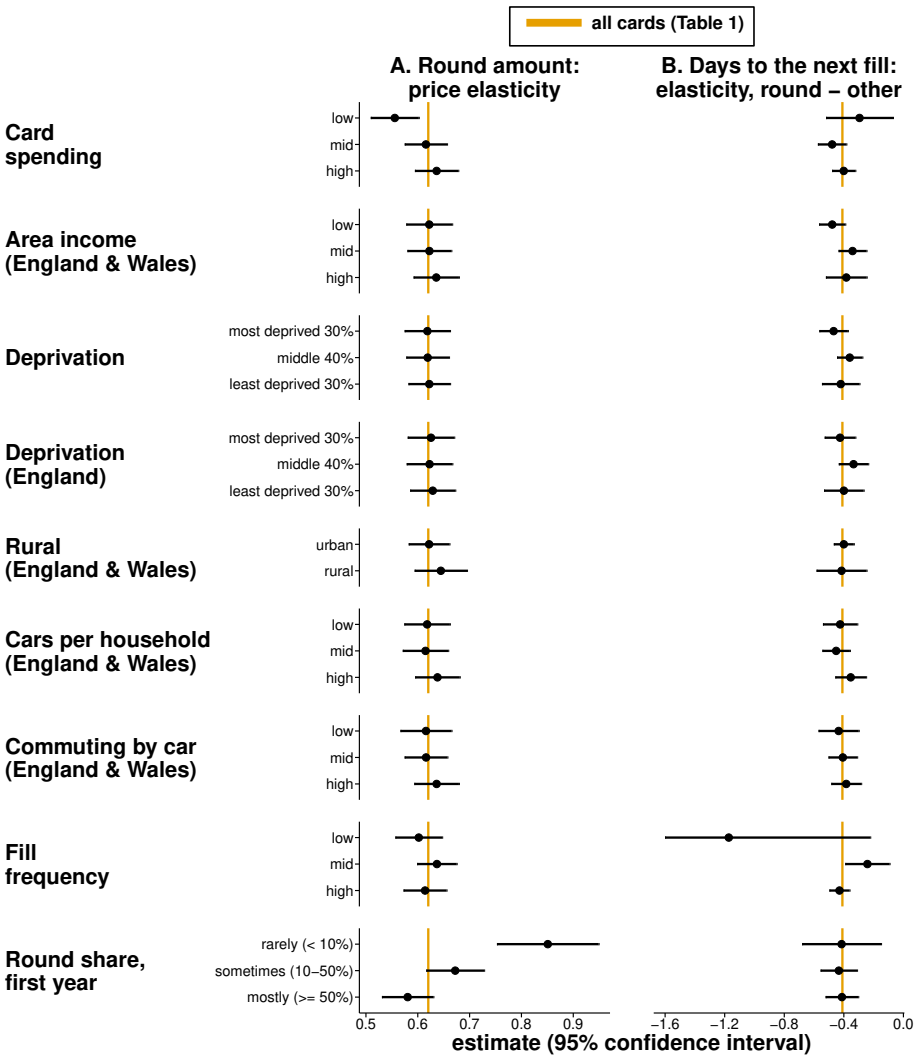
Table A3: Fill Size, Timing, and the Recent Price Change

	Change in log price, previous 8 weeks (1)	Log price (2)	Observations
Other fills: log liters	−0.08	−0.23	769,119
	(0.03)	(0.02)	
Log days to next fill: other fills	0.04	−0.24	1,303,925
	(0.10)	(0.07)	
Log days to next fill: round − other	−0.09	−0.40	1,303,925
	(0.04)	(0.03)	
Log round amount	−0.16	0.64	1,583,567
	(0.04)	(0.02)	

Notes: Fill-level models with the change in the log national daily price over the previous eight weeks added to the log price. Other fills: log liters, with log days since the previous fill as a control. Days to the next fill: equation (2) with the recent change and its interaction with the round-fill indicator. Round amount: equation (1). Card (card × amount for days to the next fill) and month-of-year fixed effects, day-of-week fixed effects (except for the round amount), trend, and COVID indicator; clustered by card and month.

D Heterogeneity

Figure A4: Estimates by Income, Car Dependence, Fill Frequency, and Habit



Notes: As Figure 3. Vertical yellow lines: pooled estimates. Group estimates come from separate regressions. Tests of differences come from one pooled model per characteristic that interacts the group with the price and shares the controls across groups. Of the 34 tests, 7 are significant at the 5 percent level and none after a Bonferroni correction. The interval for the low fill-frequency group extends beyond the left edge of Panel B and is cut there. Card spending: terciles of spending per active month (a weak income proxy, because people hold several cards). Area income: Office for National Statistics (ONS) net household income of the Middle Layer Super Output Area (England and Wales). Deprivation: within-nation deciles of each nation's index of multiple deprivation (the government's official measure of small-area deprivation, combining income, employment, health, education, crime, housing, and living environment; each nation computes its own, so the indices are not comparable across nations) and, separately, England's index alone. Car dependence (England and Wales, Census 2021): postcode sectors with mostly rural households, and terciles of cars per household (Car or Van Availability, TS045) and of the share of commuters who drive (Method Used to Travel to Workplace, TS061). Areas are matched to postcode sectors with the ONS Postcode Directory. Fill frequency: days with a fill per month with a card payment. Round share in the first year: rarely (below 10 percent), sometimes, mostly (50 percent or more), with estimates on the card's later fills.